VLSI Testing: 2009 Final

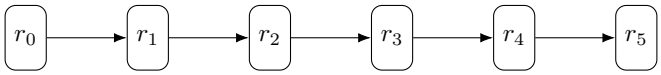
Reconstructed questions and independently derived solutions. Personal names, identifiers, handwritten answers and administrative headers are omitted. Figures are redrawn as logical diagrams or explicit netlists. A source-reading note identifies a meaningful ambiguity instead of silently changing the question. Finite Boolean checks are supplied in the examination collection's Python verification script.

1 LFSR quotient stream and signature

Question. Compute the quotient and final signature for two six-stage LFSRs, feeding the rightmost input bit first, for

$$M = \texttt{100001110110010011101010}.$$

The first circuit distributes the last-stage feedback into XOR taps; the second XORs selected register outputs into the serial input.



Solution. The drawing does not state a seed, so assume a cleared register. Number stages $r_0, \ldots, r_5$ from input to output and read the outgoing r_5 *before* each shift. No extra flush clocks are applied. All sums below are XORs over GF(2).

For the distributed-feedback circuit, let $f = r_5$. First form $(r'_0, \ldots, r'_5) = (u, r_0, r_1, r_2, r_3, r_4)$, then XOR f into positions $\{0, 3, 4, 5\}$. For the input-feedback circuit,

$$r'_0 = u + r_0 + r_1 + r_2 + r_5, \qquad r'_i = r_{i-1} \quad (1 \le i \le 5).$$

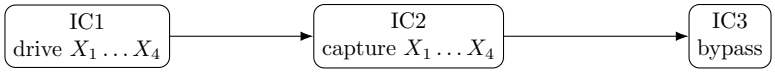
These recurrences, rather than a visually reversed polynomial label, define the bit ordering unambiguously.

Circuit	Serial output, first bit first	Final $(r_0, \ldots, r_5)$
Distributed	`000000111101000000100100`	`101001`
Input feedback	`000000111101000000100100`	`001011`

The first six outgoing bits are the initial zero state. Deleting that six-bit latency gives the emitted quotient portion during the remaining input clocks. The table gives physical register signatures; Fibonacci and distributed-feedback state encodings differ although their outgoing streams agree. Reversing the signature, clocking six extra zeros or using a nonzero seed changes the printed result and must be stated.

2 IEEE 1149.1 interconnect testing

Question. Three ICs form the serial chain $TDI \to IC1 \to IC2 \to IC3 \to TDO$. The four lines $X_1, \ldots, X_4$ run from IC1 outputs to IC2 inputs. Describe the instructions and patterns required to detect both stuck-at values of interconnects $X_1, \ldots, X_4$.



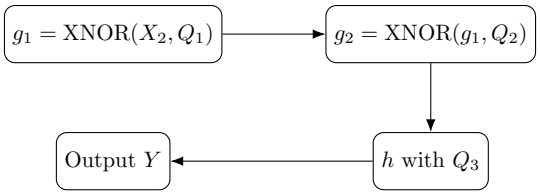
Solution. Put IC1 and IC2 in EXTEST; put the uninvolved IC3 in BYPASS. Preload safe drive values, update the launch cells, allow interconnect settling, capture at IC2, then shift out the captured response. Drive 1111 to detect SA0; drive 0000 to detect SA1. Other boundary cells must have safe values, especially output enables, to prevent contention.

1149.1 versus 1500. IEEE 1149.1 primarily gives chip/board boundary access through a TAP controller and a standardized serial state machine. IEEE 1500 wraps embedded cores inside an SoC; it separates the wrapper from the system-level test access mechanism and supports wrapper serial access with optional parallel access for bandwidth. Thus their intended integration level and access/control architecture differ. See the TI boundary-scan design note and IEEE 1500 standard overview linked in `exams/README.md`.

3 Full-scan conversion and stuck-at tests

Question. Convert the sequential circuit to full scan, derive tests covering all stuck-at faults of the combinational logic, and describe the application sequence. The reconstructed normal-mode netlist is:

Signal	Defining expression
g_1	$\overline{X_2 \oplus Q_1}$
g_2	$\overline{g_1 \oplus Q_2}$
h	$\overline{g_2 \oplus Q_3}$
Y	$X_1 + h$
D_1	g_2
D_2	h
D_3	g_1



Solution. Replace every D flip-flop by a muxed scan flip-flop: $D_i^{\text{scan}} = SE?SI_i : D_i$. Connect $SI \to FF_1 \to FF_2 \to \cdots \to FF_n \to SO$ with common scan enable and a controllable clock. Scan state outputs are independent pseudo-inputs; all functional D inputs become observable pseudo-outputs.

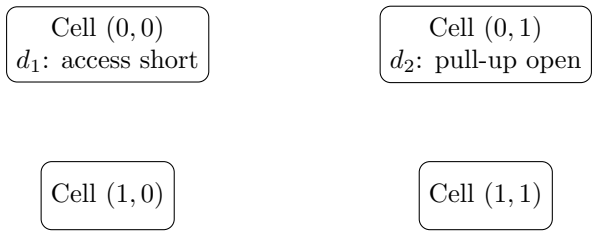
$(X_1, X_2, Q_1, \ldots, Q_n)$	$(D_1, \ldots, D_n)$	Y
00001	001	0
00101	110	1
11110	101	1

Exhaustive enumeration of all 2^5 input assignments verifies 46/46 single stuck-at faults for this netlist: both values on every signal stem and on every individual gate-input branch. Identity buffers at the observed D pins also model their separate output branches. No modeled faults are redundant. The displayed set is sufficient, not claimed minimum.

For each row: assert $SE = 1$; shift $Q_n, \ldots, Q_1$ in that order; set X_1, X_2; deassert SE and observe Y before the capture edge; apply one functional capture clock; reassert SE and unload $D_n, \ldots, D_1$ while loading the next state. Account for scan-in/out bit order and compare only the expected response positions. The proof covers combinational SAFs, not all scan-mux, clock, reset or analog defects.

4 Defects in a two-by-two SRAM

Question. A 2×2 bit-oriented SRAM has defect d_1 across a cell's access transistor and defect d_2 in a pull-up connection. Identify the faults and develop a March test for d_2. **Source-reading note.** The scan shows d_1 as a bypass/bridge across the left access device of cell $(0, 0)$, and d_2 as an open in the right pull-up path of cell $(0, 1)$. The interpretation below states this explicitly.



Solution. d_1 makes the cell permanently coupled to its bitline even when its wordline is not selected. It can cause read/write disturbance or coupling between cells sharing the column; a unique single-cell stuck-at label cannot be justified without bitline/precharge and strength assumptions. d_2 removes a restoring pull-up path. A write can initially charge the affected node, but leakage may then destroy one stored polarity: a data-retention/stuck-open fault, rather than necessarily a static SAF.

Use a retention-extended March sequence, testing both data polarities:

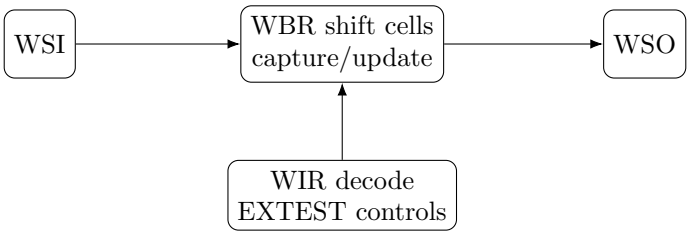
$$\{\Updownarrow(w0);\ \text{wait}(t_R);\ \Updownarrow(r0, w1);\ \text{wait}(t_R);\ \Updownarrow(r1, w0)\}.$$

Every cell must wait at least t_R after its own write and before its read. A global wait after an all-cell write guarantees that lower bound. Choose t_R from the SRAM retention specification; no value is supplied in the exam. The first read phase catches loss of 0; the second catches loss of 1. An immediate read after a write can miss the open pull-up because the node may temporarily retain its capacitive charge.

5 IEEE 1500 wrapper external test

Question. A wrapped core has two internal scan chains, a wrapper boundary register (WBR), a wrapper instruction register and a bypass register. (a) For an ExTestS serial external test, describe

the data path and application. (b) Modify the architecture to support an ExTestP parallel external test with reduced test time.



Solution. Shift the ExTestS instruction through the wrapper instruction path and update the instruction latch. Select the boundary register, bypassing core-internal scan chains for this external test. The serial test-data path is WSI→WBR cells around the core→WSO. Shift known values into source-side boundary cells, perform UpdateWR to launch them into the interconnect/glue logic, allow settling, capture receiving-side boundary responses with CaptureWR and shift them out with ShiftWR. Instruction bits are control, not inserted into every external-test data word.

For ExTestP, provide a wrapper parallel port and a TAM of width w, partition the active boundary cells into w accessible segments, and add mode-controlled input/output selection to connect each segment to a lane. Balance segment lengths, retain capture/update behavior and preserve serial access for setup and compatibility. With total N shifted bits, the ideal shift count changes from N to roughly $\lceil N/w \rceil$, subject to the longest actual segment and fixed control overhead. Simply adding an opcode without parallel data routing does not reduce time.